EmoNet: A Transformer based network for Conversational Emotion Recognition

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AGENDA

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Motivation and Background

- The Emotion Recognition in Conversation(ERC) enriches the understanding of human-machine interactions, helps advancements in various domains, including customer service, mental health, and educational technology.

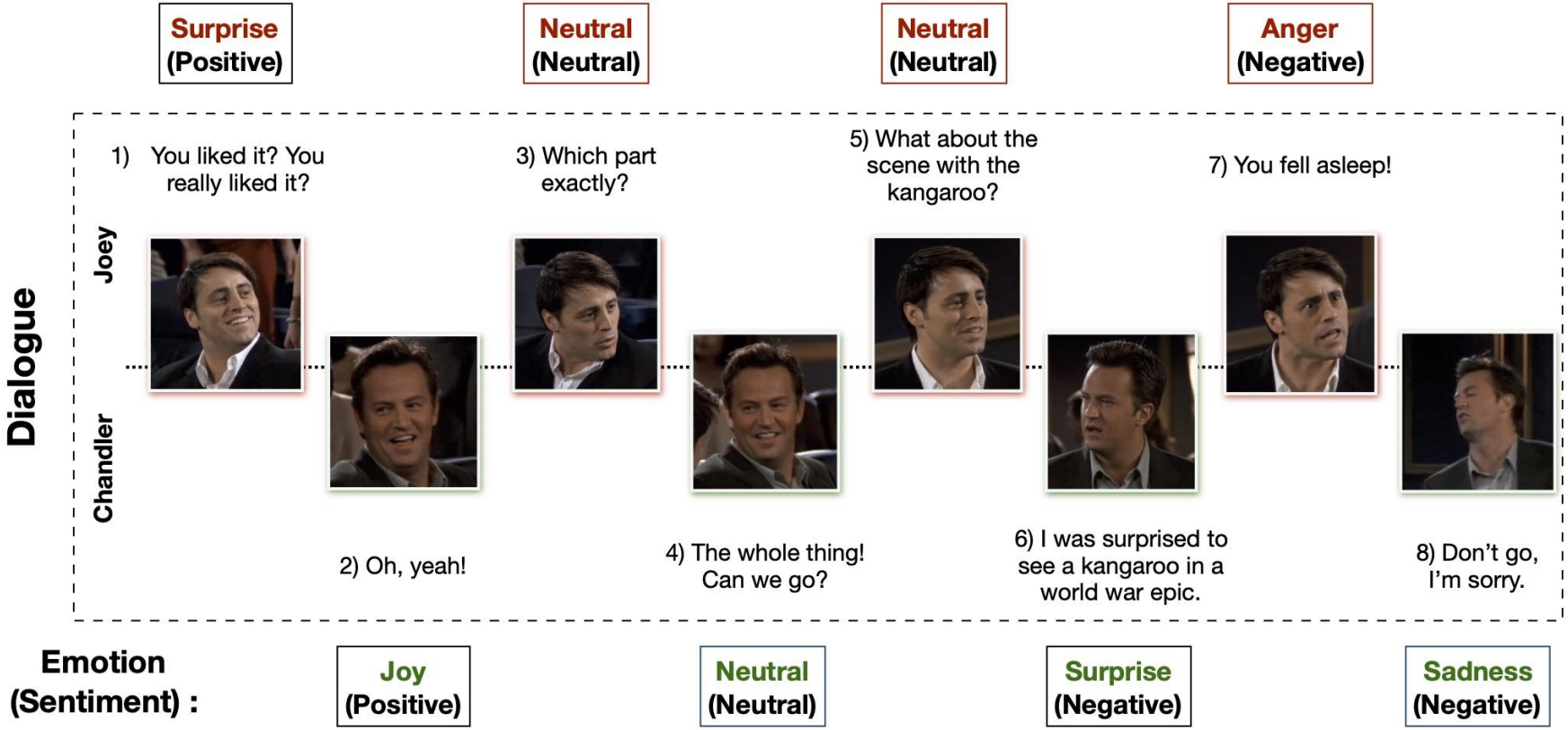
- ERC has potential to enhance the emotional intelligence of machines, thereby revolutionizing the way we interact with technology.

- The integration of ERC leap towards creating more empathetic and responsive AI systems.

- Current state-of-the-art in ERC on the EmoryNLP dataset shows a Weighted-F1 score of 41.39, indicating the existing progress and areas for improvement in the field.

ERC – What it is?

- The ERC task aims to identify the emotion of each utterance from several pre-defined emotions from the transcript of a conversation along with speaker information of each constituent utterance



Literature Review

- The current state-of-the-art ERC literature to understand architectures.
  - The architectures adopted technologies ranging from Transformer, GCN, CNN, GRU, LSTM
  - The methodologies range from commonsense knowledge, curriculum learning, hierarchical self-attention, interspeaker influence, etc
  - The models are getting complex but the predictive gain is not significant
  - The current state-of-the-art ERC lead score is 41.39
- Selected a model (CoMPM) part of the SOTA, as there is scope for further studies, and relatively simple.

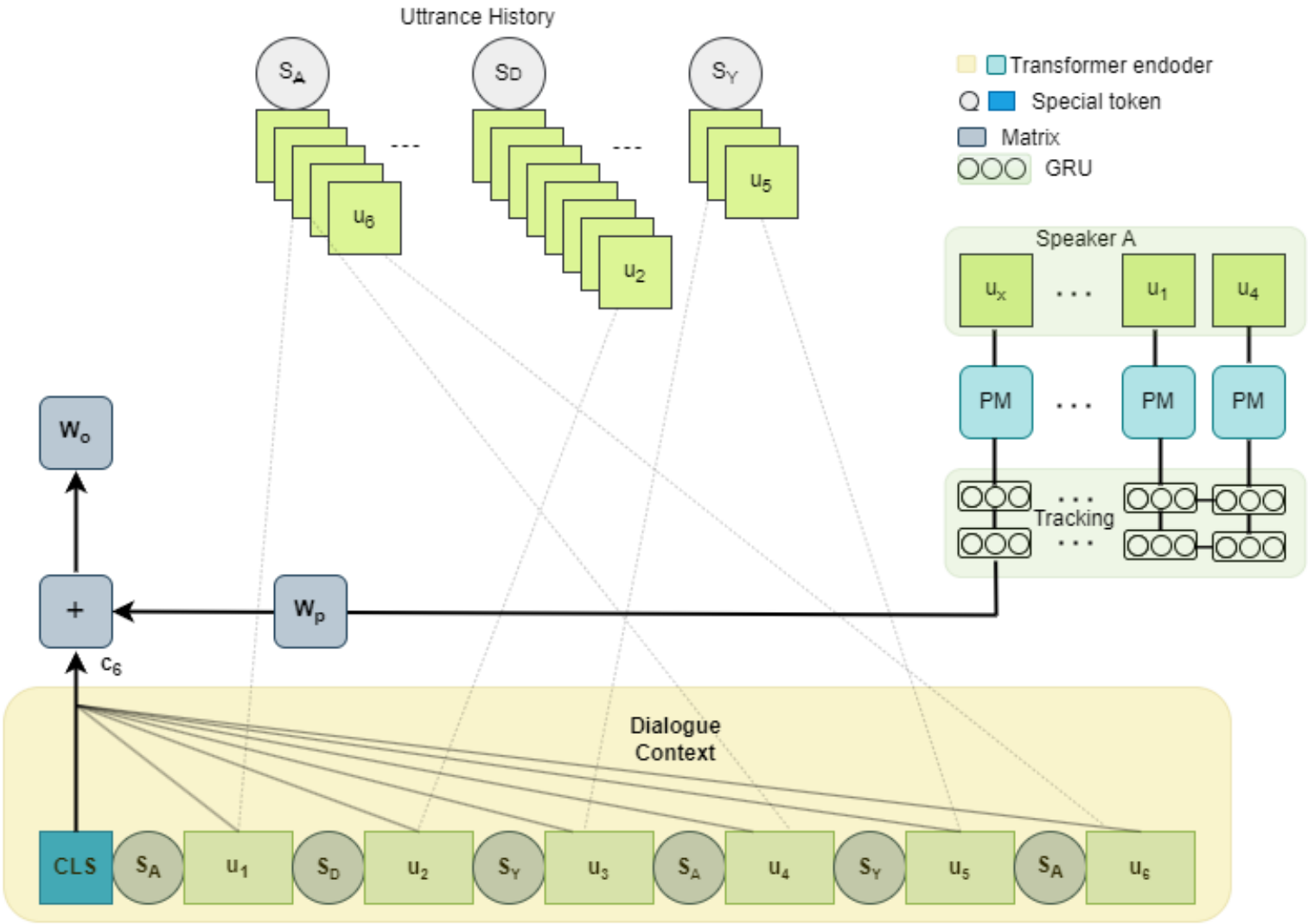
Research Questions

- How can we **improve the performance** of emotion recognition models in conversation beyond the current state-of-the-art?
- What **novel architectures** or techniques can be explored to enhance emotion recognition accuracy on conversation datasets like EmoryNLP?
- Are there specific **aspects of conversation data that existing models fail to capture** adequately, and how can we address these limitations to achieve higher accuracy?
- What are the **potential limitations or biases in the existing state-of-the-art** model for emotion recognition in conversation, and how can these be addressed?

METHODS

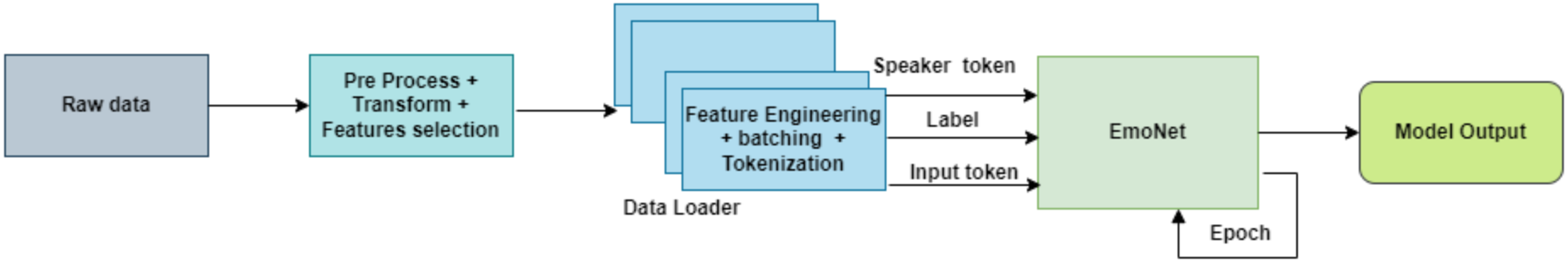
- Data
  - EmoryNLP (https://github.com/emorynlp/emotion-detection)
  - Emotion labels are primary emotions in the Feeling Wheel (Willcox, 1982).
    - Emotion labels: Neutral, Joyful, Peaceful, Powerful, Scared, Mad, and Sad
- Feature Engineering
  - Built Dialog context and Speaker Behaviour from the input features
  - Used RoBERTa tokenizer to tokenizer the features
- EmoNet (A Novel Architecture)
  - Experimented with RoBERTA & GRU models.
  - Trained on A6000 GPU with 48GB hardware on Linux VM
- Evaluation Matrix
  - Best Weighted F1 score on dev dataset is selected among the training Epochs

EMONET – A NOVEL ARCHITECTURE FOR ERC



This model consists of two modules: a Dialogue context embedding module and a Speaker Behaviour module. The figure shows an example of predicting emotion of u6, from a 6-turn dialogue context. A, D, and Y refer to the participant in the conversation, where SA = Su1 = Su4 = Su6, SD = Su2, and SY = Su3 = Su5. Wo and W_p are linear matrices

END-TO-END SOLUTION FRAMEWORK



End-to-End Solution Framework

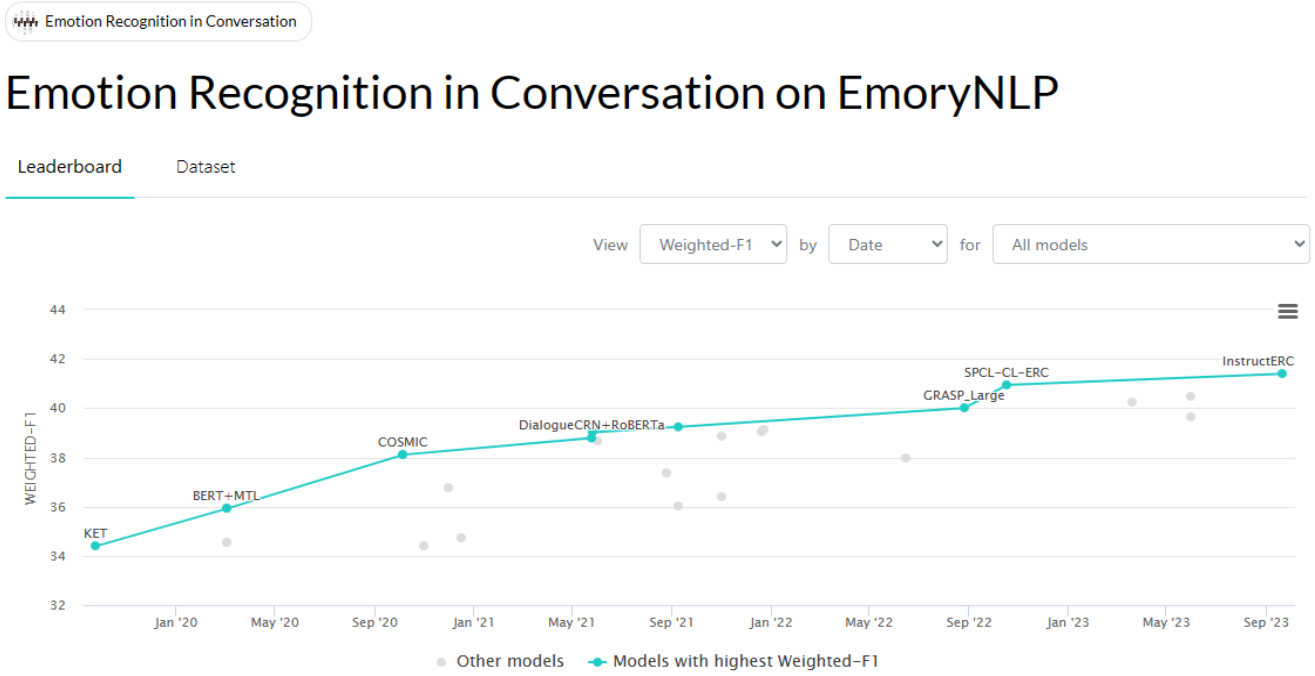
RESULTS

- Various experiments during model building

Models	Weighted F1
Base(ComPM)	37.85
+ Global Speaker ID	29.43
+ Speaker Behaviour	36.96
+ Weighted CE loss	37.13
+ Normalised Weighted CE loss	**39.18**

- The relative position of EmoNet in STOTA rank table

Model	Weighted F1	SOTA rank
InstructERC	41.39	1
TUCORE-GCN_RoBERTa	39.24	7
EmoNet	**39.18**	
S+PAGE	39.14	8
CoMPM (base for EmoNet)	37.37	16



Discussion and Conclusion

- **Multimodal Emotion Recognition**: Emotion understanding involves multiple cues like tonality, facial expressions, and body language.
- **Textual Conversation Challenges**: Text-based interactions lack non-verbal cues, limiting emotional context, seen in datasets like EmoryNLP.
- **Contextual Focus**: Current research emphasizes dialogue context, but it may not capture all emotions.
- **Behavioral Modeling**: Integrating speaker personality traits supplements emotional understanding in textual interactions.
- **Improved Recognition**: Combining contextual analysis with behavioural modelling enhances emotion recognition accuracy.
- **Future Directions**: Exploring new methods to fuse contextual and behavioural data can advance emotion recognition systems.

CONTRIBUTIONS

This research introduced a **novel architecture for ERC** on textual data. Below are the contributions of knowledge

- ❑ We propose that the **speaker needs global identity** across all data the model is exposed to rather than the dialogue level identity. This helps the prediction based on past knowledge.
- ❑ We model **Speaker Behaviour** based on historical utterances. Also, propose a method of limiting computing and memory needs in case the historical utterances grow big.
- ❑ We propose to use **Weighted Cross Entropy loss** to deal with class imbalance in the ERC dataset rather than using any data augmentation and sampling technique

FUTURE RECOMMENDATIONS

The availability of **Global speaker identity and speaker utterance history** has opened up new possibilities to investigate. These possibilities are related to:

- ❑ Efficient model for participants' behaviour patterns.
- ❑ Interworking the behaviour of different participants and its influence on the emotion of the current speaker

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KIITOS
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KIA ORA
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